

Maximizing Profit: A Strategic Plan to Increase Table Sales by 20% in the Philippines

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Executive summary

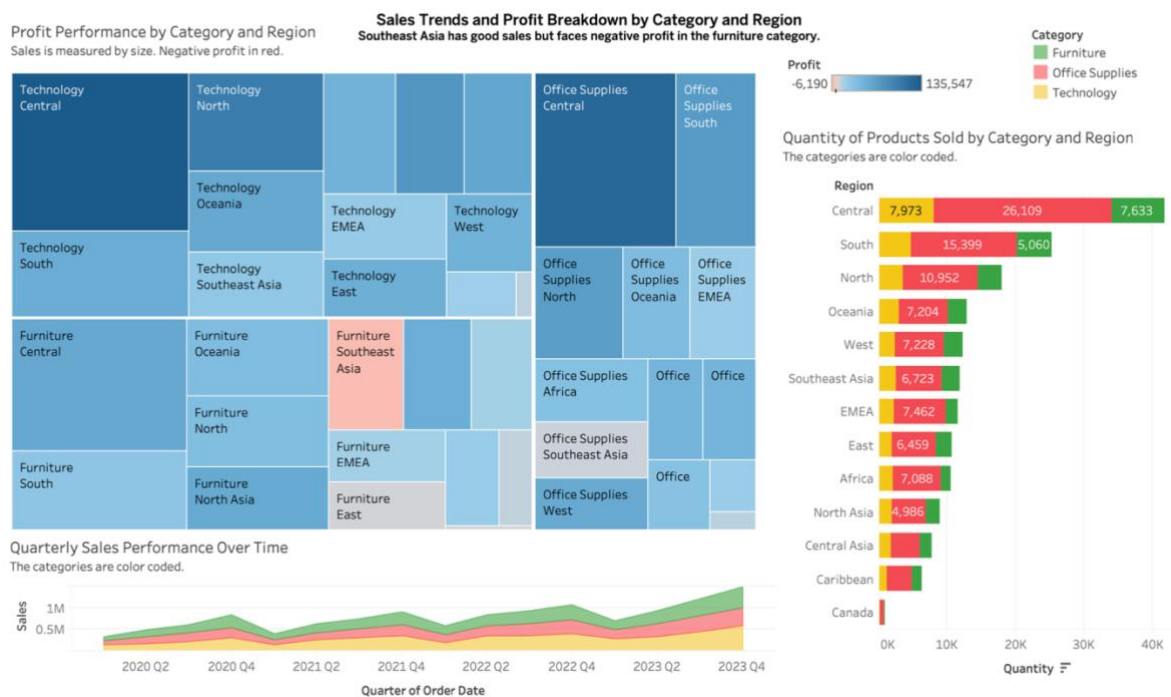
Balancing cost efficiency, competitive pricing, and sustainable growth makes profit maximization a constant challenge. Maximising profit as a firm allows the business to reinvest in their operations, expand their offerings and increase their competitive edge. According to my analysis, the “Tables” sales in Philippines exhibit the lowest discount-to-sales ratio and have the lowest profit margin. A low discount-to-sales ratio refers to a situation where the amount of discount offered on the product is relatively small compared to the total sales revenue. While this maintains higher prices, it could deter consumers, and when combined with high competition, it may lower the overall profitability. Offering substantial discount, stimulate demand by encouraging new consumers to try out the product and simultaneously attracts current consumers to purchase more. My findings indicate that raising the current discount amount by 0.2 will lead to a 20% increase in predicted sales and successfully increasing profits. However, a strategic balance is necessary to ensure that discounting drives volume growth without significantly eroding profit margins. Implementing this strategy, along with ongoing performance monitoring, can enhance profitability and long-term sustainability.

1.0 Introduction

The data analysis is based on a sample dataset of a superstore, aiming to evaluate sales and profit performance. The dataset includes information about orders recorded in the system, it can be categorised into several categories, from geographic data, product details, and numerical data. Additionally, it contains customer-related information, like customer name and ID. The main objective is to identify sales trends, and uncover potential issues affecting profitability. By interpreting these factors, the findings will provide actionable insights to help minimize losses and optimize business strategies for sustainable growth.

2.0 Superstore Performance Overview

Figure 1:



The performance of the superstore is assessed based on key metrics such as sales, profit, and market trends. This overview highlights strengths, areas for improvement, and potential growth opportunities. The main takeaway is that Southeast Asia has good sales but faces negative profit in the furniture category.

2.1 Profit Performance

A heat map is used to visually represent profit and sales performance across different categories and regions. The highest profit and sales are recorded in the Technology category

in the Central region, making \$135,547 in profit and \$1,038,206 in sales. In contrast, Southeast Asia's Furniture category shows the lowest profit at -\$6,109, despite having substantial sales of \$310,954. This suggests potential issues such as high costs, pricing inefficiencies, or low discounts impacting profitability. Profit is indicated by colour intensity based on the reference scale and the sales is represented by the box size.

2.2 Seasonal Sales Performance

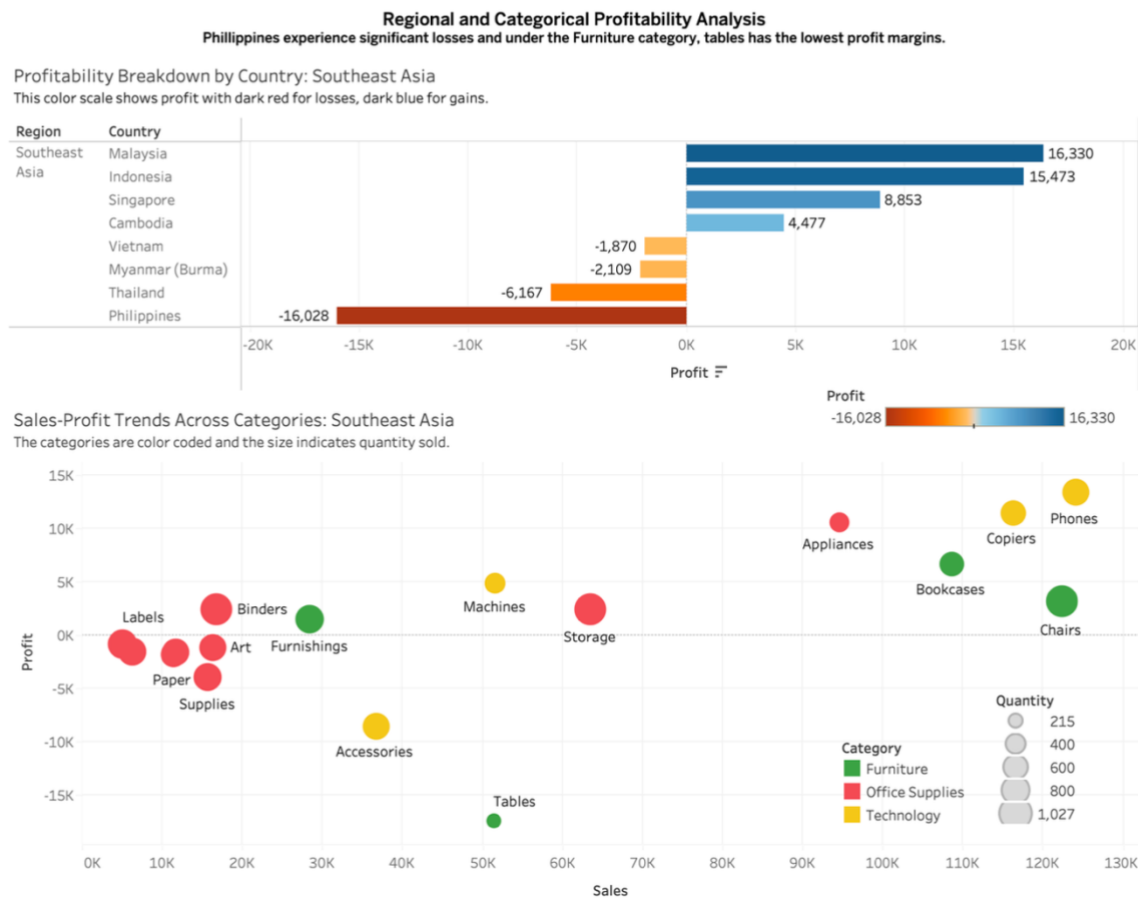
Next, a filled line graph illustrating categorical sales trends over a four-year period (2020–2023). The overall trend show a steady increase in sales, with all categories performing similarly. A noticeable dip occurs in the first quarter of each year, likely due to seasonal demand fluctuations. In contrast, sales peak significantly in the fourth quarter, which may be caused by holiday shopping, or end-of-year business purchases.

2.3 Product Performance

A bar graph is used to show the quantity of products sold across different categories and regions. The categories are represented by three distinct colours: green for Furniture, red for Office Supplies, and yellow for Technology. The Central region generated the greatest quantity across all the categories, with 7,973 units in Technology, 26,109 units in Office Supplies and 7,633 units in Furniture. Overall, Office Supplies contributed the largest number of products, suggesting higher demand or lower price points compared to other categories.

3.0 Regional and Categorical Profitability Analysis in Southeast Asia

Figure 2:



From the previous dashboard analysis, Southeast Asia was the only region facing negative profit, prompting a deeper analysis. This analysis examines which countries in Southeast Asia and product categories are contributing to the negative profit. The key takeaway is that Philippines is experiencing significant losses, and under the Furniture category, “Tables” has the lowest profit margins, highlighting potential areas for strategic intervention to improve profitability.

3.1 Profitability Breakdown

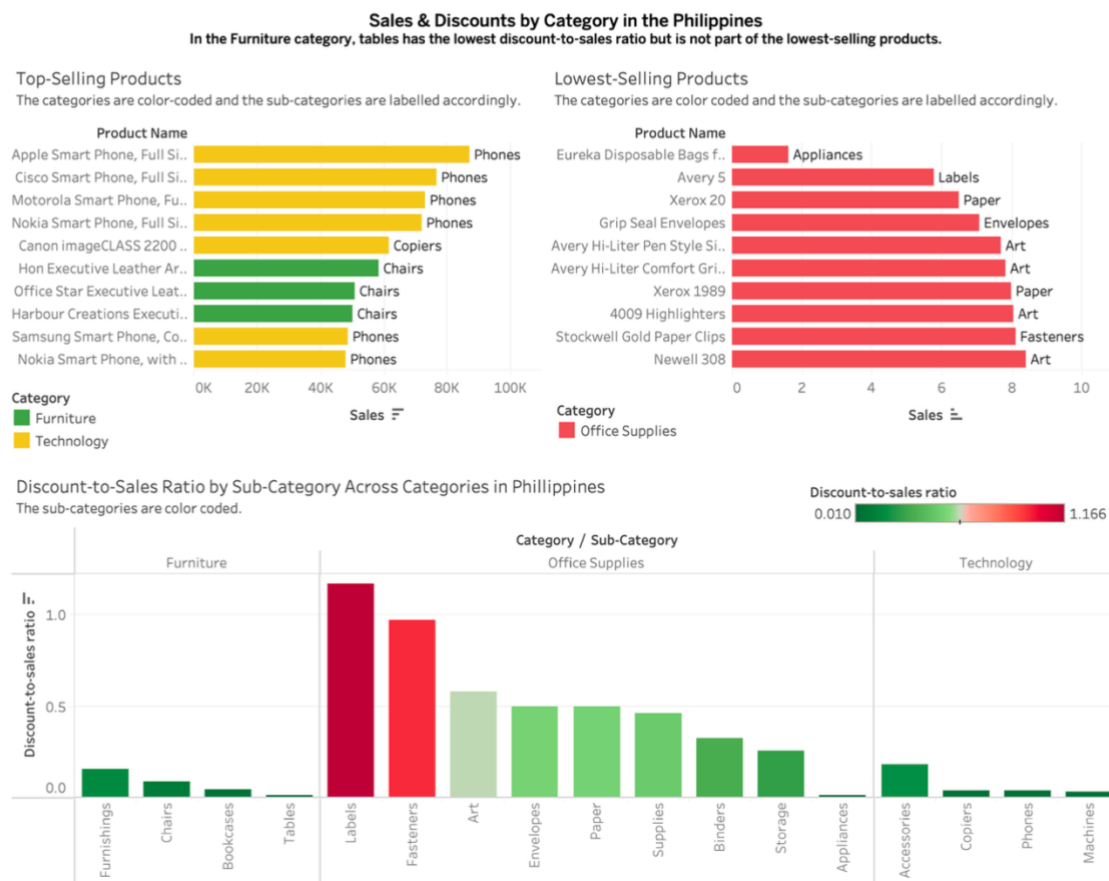
Firstly, a bar graph is used to depict the profit generated by each country in Southeast Asia, with a colour intensity scale applied for better visualization—gains are shown in blue, while losses appear in red. Malaysia recorded the highest profit at \$16,330 while Philippines experienced a significant loss of -\$16,028, highlighting a critical area of concern for profitability.

3.2 Sales-Profit Trends Across Categories

Next, a scatter plot visualising sales and profit trends across different product categories in Southeast Asia, with categories distinguished by colour and the size of each point representing the quantity sold. The x-axis shows sales, while the y-axis displays profit, showing a clear distinction between profitable and unprofitable categories. Particularly, "Chairs" and "Phones" (in green and yellow, respectively) have high sales and positive profit, while "Tables" (in green) exhibit significant losses despite substantial sales. Overall, this chart showcases the varying financial performance of different product categories within the region.

4.0 Sales and Discounts Breakdown in the Philippines

Figure 3:



From the previous dashboard analysis, the Philippines experience significant losses, sparking in-depth evaluation. This breakdown accesses which categories are causing the significant losses and if discounts has a part to play. A notable insight is that in the Furniture category,

“Tables” has the lowest discount-to-sales ratio, yet it is not part of the lowest selling product overall.

4.1 Top and Lowest Selling products

A bar graph is used to illustrate the top 10 and bottom 10 sellers, the categories are colour-coded as indicated above and the sub-categories are clearly labelled. The Top-Selling Products table includes only items from the Furniture and Technology categories.

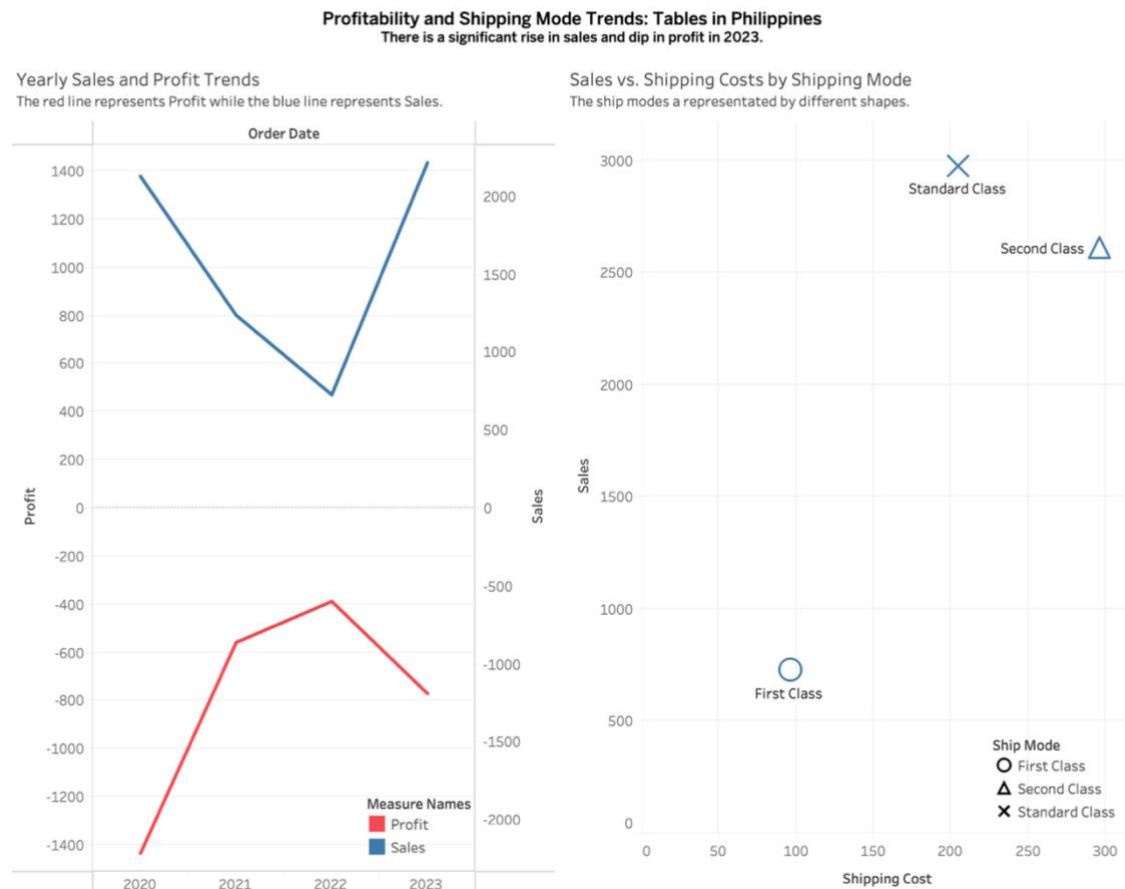
Interestingly, despite Office Supplies having the highest quantity of products listed in the Section 2.3, this category appears solely in the Lowest-Selling Products table. This discrepancy could be due to the typically lower price points and shorter product lifespans of office supplies, in contrast to the higher-value and longer-lasting furniture and technology products.

4.2 Discount-to-Sales Ratio Analysis

Next, we used a bar graph to visualise the discount-to-sales ratio trends across different product categories in the Philippines, a colour intensity scale applied for better visualization—high ratios in red and low ratios in green. The discount-to-sales ratio is calculated by dividing the total discounts by total sales, helping us assess whether the applied discounts are effectively driving sales. Within this data, “Labels” in the Office Supplies category has the highest ratio at 1.166 and “Tables” in the Furniture category has the lowest ratio at 0.010. As highlighted above in the Section 3.2, “Tables” experienced substantial sales. A low discount-to-sales ratio, combined with high sales and low profit, suggests that most sales occur with minimal discounts. This could indicate that, while the company is successfully moving a large volume of products, the profit margin per item is low. As a result, they are not retaining a significant portion of their sales revenue as profit. This suggest potential issues like high operational costs, inefficient pricing strategies, or both.

5.0 Profitability and Shipping Mode Trends in The Philippines

Figure 4:



Referring to the previous dashboard analysis, high operational costs, inefficient pricing strategies or both could be contributing to the low profit margins for “Tables” in the Philippines. In this dashboard analysis, we will be taking a look at the high operational costs, with an emphasis on yearly profit and sales trends, as well as shipping mode trends. An essential takeaway is that the shipping mode does not appear to play a significant role in causing the low profit margins. However, the trends in sales and profit suggest that there may be other underlying factors that are more impactful.

5.1 Yearly Sales and Profit Trends

Firstly, we have a line graph to show the sales and profit trends over 4 years, from 2020 to 2023. The red line represents Profit while the blue line represents Sales. Notably, a trend is particularly evident in the “Tables” product category, where sales and profit appear to follow an inverse proportional relationship. The inverse proportional relationship between sales

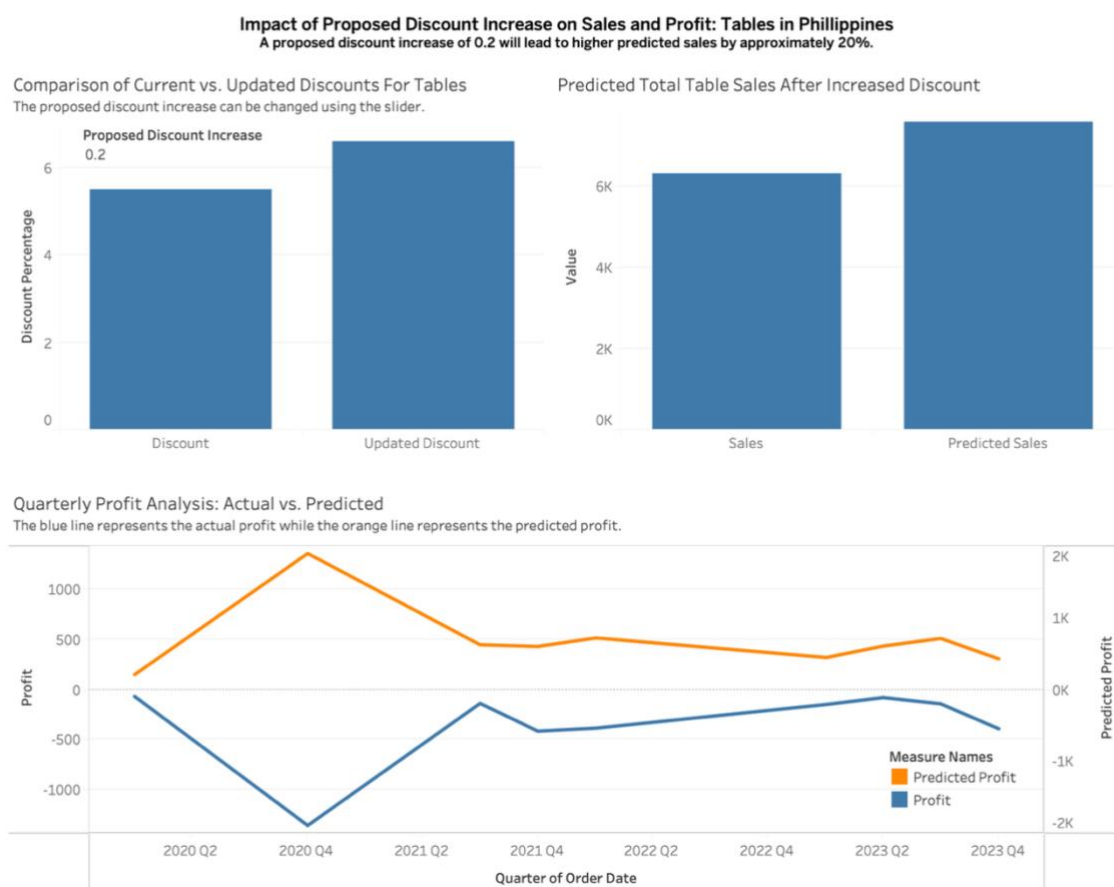
and profit typically arises when an increase in sales is coupled with a reduction in profit margin (due to discounts, lower-priced products, or increased operational costs) .

5.2 Shipping Mode Trends

Since high operational costs may be the reason for low profit margins, we analysed shipping costs to determine if they are a significant factor. A scatter plot visualizing sales and shipping costs by shipping mode was used, with different symbols representing each mode. Standard Class has the highest sales (~3000) but also the highest shipping cost (~250). First Class has the lowest sales (~800) but also the lowest shipping cost (~100). There is no strong correlation between shipping costs and sales, indicating that shipping expenses are not the primary driver of low profit margins. Other factors, such as product pricing, discount strategies may play a more significant role.

6.0 Proposed Solution

Figure 5:



From the previous analysis, we have concluded that “Tables” category in the Philippines is experiencing low profit margins. Our findings indicate that operational costs, such as shipping expenses, are not the primary factor. Instead, the low discount-to-sales ratio, as mentioned in Section 4.2, appears to be a key driver of reduced profitability. Hence, in this dashboard, we will adjust the discount levels for "Tables" to analyse their impact on total sales and determine if optimizing discounts can improve profit margins.

6.1 How an Increased Discount Will Affect The Total Sales

A proposed discount increase of 0.2 has been applied to a bar graph to visualise the increase. The discount percentage has increased 5.5% to 6.6%. Additionally, another bar graph has been created to illustrate how this increased discount is predicted to affect the total sales. According to the data, sales of “Tables” are projected to increase from \$6,309 to \$7,570, representing an increase of approximately 20%.

6.2 Impact on Predicted Profit

As mentioned earlier in Section 5.1, sales and profit typically follow an inverse proportional relationship. However, the line graph comparing the actual and predicted profit trend reveals a significant increase in profit, shifting from negative to positive. This suggests that increasing the discount on "Tables" in the Philippines leads to higher profitability.

7.0 Conclusion

In conclusion, we analysed the reason for low profitability in the Southeast Asia region. First, from the performance analysis we realise that Furniture in the Southeast Asia region is facing negative profit. Secondly, we found that Philippines experience significant losses and “Tables” has the lowest profit margins. A deeper examination of the Philippines indicated that "Tables" had the lowest discount-to-sales ratio despite generating substantial sales. Subsequently, we ruled out operational cost as a primary driver of the low profit margin. Ultimately, we determined that adjusting the discount rate would help increase profitability, leading to improved business performance.